

Neural Computing and Applications

What Can a Swiped Word Tell Us More? Demographic and Behavioral Correlates from Shape-Writing Text Entry --Manuscript Draft--

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Abstract:	Shape-writing (aka gesture typing or swiping) is a word-based text entry method for touchscreen keyboards. It works by landing the finger on (or close to) the first character of the desired word and then sliding over all the other character keys without lifting the finger until the last word character is reached. This generates a trajectory of swiped characters on the keyboard layout which can be translated to a meaningful word by a statistical decoder. We hypothesize that swiping carries rich information about the user, such as demographic (e.g. age or gender) and behavioral (e.g. swiping familiarity or input finger) information. To test our hypothesis, we trained several sequence classifiers using different recurrent neural network architectures to predict demographic and behavioral correlates of users from swipe trajectories. We show that our sequence classifiers are always performing better than a random classifier, therefore we conclude that cognitive and motor control mechanisms are embodied and reflected in swipe trajectories, validating thus our research hypothesis. Taken together, our results have implications for user privacy. Currently swiping is supported by all mobile vendors and has millions of users, so people may be inadvertently profiled at an unprecedented granularity. Future work should consider new ways of addressing these issues without impacting the user's swiping experience.	
Response to Reviewers:	Dear Editor and Reviewers, We thank you for the timely and insightful feedback provided. In the following we describe how we have addressed the raised issues and, as a consequence, have revised our manuscript accordingly. We have used orange font color to highlight the incorporated changes.	

=== Reviewer #2 ===

> I would rather see fewer targets with a more balanced data set. The balance here is one factor I find most difficult to get past. Given the number of target classes, there are 1152 possible combinations, not all of which may be represented, and are certainly not represented equally.

Thanks for the suggestion. We have mentioned this exercise as future work in Section 5. We believe it is more fundamental and necessary to study the original swiping dataset in a systematic way, in order to get a more realistic estimate of the performance of our models.

> They do not consider any interactions, such as gender and age, and do not adjust for these. So how about limiting the dataset to RH females? You mention that the dataset is skewed towards this group.

This is very interesting, many thanks again for the suggestion. We have mentioned it in Section 5, where we discuss the limitations and future work.

> Page 8 says 61% are non-US, but Pg 19 states that the majority are young female RH US citizens, so I wonder even more about the specific data distribution.

Actually, the group of “young female right-handed participants” are mostly from the US. Since the “US” bit in that sentence (on page 19) was confusing, we have removed it.

> Continuing with the 39% who are US citizens, how do we know their primarily spoken language is English? It might be, say, Spanish if they got citizenship as adults. So perhaps a more important identifier is “what language do you speak primarily?”

We mentioned in Section 3.1 that language is “browser language”. According to the public dataset we analyzed, the primary language of participants was automatically inferred from their browser: it was assumed that their browser would be in the language they were most comfortable with. Please see reference [4] for more details.

> As for nationality, why is a non-US citizen using a standard English phone and keyboard?

This was a requirement for the data collection. As mentioned in Section 3.1, all participants used the same JavaScript-based virtual QWERTY soft keyboard. Please see reference [4] for more details.

> Did all the other nationalities use the same alphabet?

Yes, since the soft keyboard was the same for all participants [4].

> Why were non-English speakers writing in English? Friends of mine who are not native to the US often use a keyboard & language that fits better with their spoken language.

Because the prompted phrases were all in English [4]. We should remark that we did not collect the “How We Swipe” dataset. It was released in 2020 for benchmarking mobile-based studies.

> Does typing on an English keyboard if you are writing, say, Greek, slow you down?

This is an interesting question. Unfortunately, we do not have such an insight. We may refer to the work of Franco-Salvador et al. [2018] where they reported that, on average, Russian has more words per phrase and German has more letters per word than other Indo-European languages. Please see <https://www.sciencedirect.com/science/article/pii/S1071581918300399>

> Would the device be an important variable in speed? Blackberry vs iPhone vs iPad? Type of device, age of device, software version (latter versions guess the word).

The “How We Swipe” paper [4] reported that iPhone participants were a bit faster than Android participants, and hypothesized that it would be attributed to the device’s screen (retina vs. non-retina displays).

> Pg 11, line 60. Was this data randomly assigned to these sets, or was the data balanced on a characteristic?

We have clarified in Section 3.4 that the data splits were randomly created.

> Did every problem use the same data sets? Or did you balance on target classes within a dataset when testing that target? I know you state on page 12 that the class distributions are imbalanced for most of the targets.

Yes, each model was trained on the very same data partitions. We have clarified it in Section 3.4.

> Are reported results just on the validation set? How well did the models do on the training set?

All results are reported on the test set, as usual in Machine Learning experiments. We have clarified in Section 3.4 that the testing partition is held out exclusively for model evaluation, as it represents unseen data and is therefore a more challenging but also a more realistic scenario.

> Were the models robust or was there falloff?

Our experiments show that each target has a preferred model (see Table 1), however the 2-layered bidirectional LSTM or GRU model with 100 units was the most frequent winning architecture. So we can conclude that this model architecture is the most general one overall. Since we already mentioned in Section 6 that the 2-layered bidirectional LSTM or GRU model was preferred, we have added there the number of preferred hidden units as well.

> When you report model accuracy, what data set are you referring to?

We always refer to the same dataset (How We Swipe) and always the testing partition (a random 20% split of the dataset). We have clarified it in Section 3.4.

> For each model, could you show the classification matrix for each so we can see how many were classified as each category by the model? This helps understand the model accuracy when the data is not balanced. For example, your data set is 69% female and is 68.5% accurate in predicting gender. A classification matrix would show us whether your model is just predicting most users are female.

We have included the confusion matrices for each of the models shown in Table 1. We also have added a new paragraph discussing the results. Indeed, the confusion matrices show that the models were unable to properly predict three (out of eight) targets: gender, dominant hand, and swiping finger. In those cases, the models tended to predict the majority class.

> Title: Should be What More Can a Swiped Word Tell Us? Not: What Can a Swiped Word Tell Us More?

The title is actually a homage to the seminal paper “What can a mouse cursor tell us more” [22], therefore we want to keep it as it is.

> Pg 6, line 58: the dataset URL should not have a / at the end. As they have typed it, it does not go anywhere.

We have just tested the link as it appears in the PDF and confirm that it works.

=== Reviewer #3 ===

> I think the "both hands" cannot be combined with "left hand" in swipe hand level because the characteristics of "both hands" is very different from "left hand".

Agreed. However, as explained on page 8, eventually we ended up with a binary classification task in that case: "right hand" vs "other hand". This was so due to overcoming the class imbalance issue that appears when considering 3 classes: 74% "right hand", 21% "both hands", and 5% "left hand".

> What is G_i in equation (3)? Is it G_u ?

G_i refers to the input gate of the LSTM cell. Actually G_u is wrong in (6), as it should be G_i . We have corrected this typo in the revised version. Please note that G_u only applies to GRU cells.

> Give a full terminology for M and SD in the line 55 of the page 12 (first paragraph of the section 3.4).

Done.

> The AUC value should be over the 50% at least. 50% means that the classifier has no ability to differentiate between two class. However, some AUC in the figure 5 shows under the 50%. Is this correct?

We observed this behavior only when predicting nationality (Fig. 8d) and with $LR > 10^{-3}$. This means that the models were underfitted, i.e. they did not learn properly. This is in fact attributed to the high learning rates. Please see a follow-up discussion in this regard about the Robbins-Monro condition, directed at Reviewer #4 below.

=== Reviewer #4 ===

> The model could be modified with Attention mechanism, so that 'more affective learning process could be achieved. This can be considered as a future work.

Many thanks for the suggestion. Indeed, we have mentioned it in Section 5, where we discuss the limitations and future work.

> It is mentioned that low learning rate makes training process slow. But you have demonstrated your experiments with low learning rate only.

As reported in Section 3.4.1, we experimented with learning rates 10^{-1} , 10^{-2} , 10^{-3} , 10^{-4} and 10^{-5} , 5 variations in total. A learning rate higher than 10^{-1} (i.e. 0.1) is unlikely to work because of the Robbins-Monro condition [Ranganath, 2014]. Moreover, when $LR > 1$, stochastic gradient descent is not guaranteed to converge to a minimum. We have cited [Ranganath, 2014] in the revised version of the manuscript to better justify our choice of LR, in Section 3.4.1.

* R. Ranganath et al. Black Box Variational Inference. In Proc. AISTATS, 2014.

In conclusion, we would like to thank reviewers for their many useful comments and suggestions, which we have used to improve our manuscript. We believe that the current revision is stronger as a result, and we are confident that it will represent an excellent contribution to the NCAA journal.

We look forward to hearing from you.

Sincerely,
The authors

What Can a Swiped Word Tell Us More?

Demographic and Behavioral Correlates from Shape-Writing Text Entry

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Abstract

Shape-writing (aka gesture typing or swiping) is a word-based text entry method for touchscreen keyboards. It works by landing the finger on (or close to) the first character of the desired word and then sliding over all the other character keys without lifting the finger until the last word character is reached. This generates a trajectory of swiped characters on the keyboard layout which can be translated to a meaningful word by a statistical decoder. We hypothesize that swiping carries rich information about the user, such as demographic (e.g. age or gender) and behavioral (e.g. swiping familiarity or input finger) information. To test our hypothesis, we trained several sequence classifiers using different recurrent neural network architectures to predict demographic and behavioral correlates of users from swipe trajectories. We show that our sequence classifiers are always performing better than a random classifier, therefore we conclude that cognitive and motor control mechanisms are embodied and reflected in swipe trajectories, validating thus our research hypothesis. Taken together, our results have implications for user privacy. Currently swiping is supported by all mobile vendors and has millions of users, so people may be inadvertently profiled at an unprecedented granularity. Future work should consider new ways of addressing these issues without impacting the user’s swiping experience.

Keywords: Shape-writing; Gesture Typing; Biometrics; Neural Networks

1 Introduction

Shape-writing, also known as gesture typing, swype input, swipe to text, or just *swiping* (for short), is a prevalent mobile text entry method currently supported by all mobile vendors. Contrary to regular touch typing, where the user touches one key at a time and lifts up the finger to enter one character, swiping is a *word*-based text entry method: The finger lands on (or close to) the first key of the desired word and then, without lifting the finger from the keyboard, it traverses (the vicinity of) all the keys until reaching the last character of the word, generating a trajectory of touch points as a result. See [Figure 1](#) for some examples.

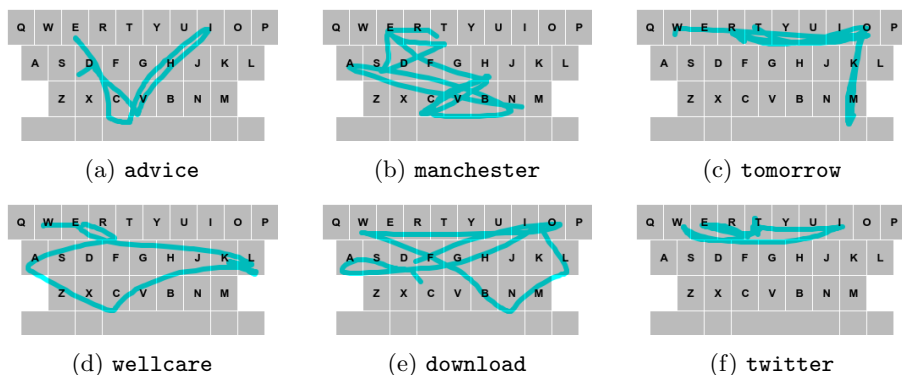


Fig. 1: Examples of swiped word trajectories.

Swiping performance has been researched in small-scale experiments [1], in large-scale studies as one of many text methods [2], or using proprietary, undisclosed datasets [3]. Collecting such data is challenging, because most mobile keyboards are vendor-locked and do not offer an API for collecting such data [4].

Until now, there was no public swiping dataset where researchers could access raw movements dynamics such as the gesture path drawn on top of the keyboard or the time lapsed between consecutive swiped keys. Recently, Leiva et al. [4] released “[How We Swipe](#)”, the largest dataset (over 1,300 participants) for conducting research on mobile swiping together with first observations of correlates of typing performance. Large-scale analyses of mobile interaction are relatively rare and mostly undertaken by commercial organizations and mobile vendors that ship soft keyboards. There are notable exceptions [2, 5–7] but they did not investigate swiping behavior.

In this work we analyse the dynamics of swiping trajectories to model and predict different demographic and behavioral correlates, including age, gender, nationality, English level, swiping familiarity, swiping hand, swiping finger, and

dominant hand. In total, we trained 200+ different classifiers with 30 different hyperparameter variations for each predicted target. We found that swipe trajectories actually carry information about demographic and behavioral correlates. To the best of our knowledge, we are the first to perform this analysis in this kind of data.

The rest of this article is organized as follows. In [Section 2](#), we present a comprehensive overview of related works in shape writing and user profiling from demographic and behavioral movement data. In [Section 3](#), we present the dataset we have analyzed, the methodology we have followed, and the results of our experiments. In [Section 5](#), we present a comprehensive summary highlighting limitations and future research directions. Finally, [Section 6](#) closes this article with a summary of the main findings.

2 Related Work

Invented by Zhai and Kristensson in 2003 [8], swiping has become a widely adopted text entry method on mobile devices. It well suits touch-based interaction, and provides a high typing speed since it relaxes the requirement of precisely acquiring each (small) key on a soft keyboard. To date, this text entry method is supported by major commercial soft keyboards including Google’s Gboard, Microsoft’s SwiftKey, and iOS’s built-in keyboard on iPhones.

The research community has carried out a large amount of research on swiping techniques. For example, swiping has been extended to support mid-air text entry [9], eyes-free input [10], ring-based input [11], phone-tilting based input [12]. In addition to text entry, swiping has also been extended to support command input [13–15], by gesturing a shortcut related to the command name. Such a method shows advantages in learnability compared with hotkey-based command input [15].

There is a large body of work that demonstrates that our digital footprints, including e.g. the websites we visit or the text we enter in social media, may help derive personally identifiable information like gender, age, location, or even political orientation. Existing literature related to online privacy provides insights around topics such as information leakage while surfing the web using desktop computers [16] or mobile devices [17]. In the following we discuss related work aimed at exploiting this information in the context of human movement data, such as handwriting and gesturing, since they are the closest work to swiping.

2.1 Biometrics Prediction from Movement Data

Leiva et al. [18] showed that accurate detection of human movements has more to do with *how* users write, rather than *what* they write. They concluded that it is better to use recurrent models than convolutional models for biometric classification, as recurrent models capture the inherent movement dynamics. However, they only explored one shallow architecture, based on Bidirectional Gated Recurrent Unit (BiGRU) cells, in the context of binary classification

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of human vs. fake handwriting. Therefore, it is unclear what design choices over the different hyperparameters (e.g. number of hidden layers, type of recurrent cell, number of hidden units, learning rates, etc.) may affect classification performance. In this work we perform an exhaustive investigation in this regard.

Leiva et al. [19] showed that our mouse movements may reveal who we are. They developed a computational model based on BiGRU cells and were able to classify the age and gender of the users with reasonable precision (near 70%) using a few lines of code. Recent work by White et al. [20] and Gajos et al. [21] could detect neurodegenerative disorders from mouse cursor movements, showing how our “digital phenotypes” could be used as adjunctive screening tools.

Chen et al. [22] were the first to notice a strong relationship between gaze position and cursor position during web browsing. Mueller and Lockerd [23] investigated the use of mouse tracking to visualize and (manually) infer the users’ interests. Since then, researchers have noted the utility of mouse cursor analysis as a low-cost and scalable proxy of eye gaze [24–26]. Several works have investigated closely the utility of mouse cursor data in web search [27–29] and web page usability evaluation [30–32], two of the most prominent use cases of this technology. Mouse biometrics is another active research area that has recently shown how to identify an individual by analyzing their mouse movements in controlled settings [33, 34]. This is the research line we focus on in this work.

Objective measurements of attentional processes are increasingly being used to explain or predict user behavior. A mouse click is often preceded by several interactions such as scrolling, hovers, movements, etc. and thus can lead to a better overall understanding of the user’s thought process. In what follows, we review research efforts that have focused on mouse cursor analysis to infer user interest, visual attention, emotions, and demographic variables like gender or age, on a desktop setting. We hypothesize that swiped word trajectories carry the same rich information about the user.

2.2 Inferring User Interest

For a long time, commercial search engines have been interested in how users interact with Search Engine Result Pages (SERPs), to anticipate better placement and allocation of ads in sponsored search or to optimize the content layout. Early work considered simple, coarse-grained features derived from mouse cursor data to be surrogate measurements of user interest [35, 36]. Follow-up research transitioned to more fine-grained mouse cursor features [37, 38] that were shown to be more effective. These approaches have been directed at predicting open-ended tasks like search success [39] or search satisfaction [29]. In a similar vein, Huang et al. [24, 40] modeled mouse cursor interactions and extended click models to compute more accurate relevance judgements of search results. Mouse cursor position is mostly aligned to eye

gaze, especially on SERPs [41, 42], and that can be used as a good proxy for predicting good and bad abandonment [43].

2.3 Inferring Visual Attention

Mouse cursor tracking has been also used to survey the visual focus of users in sponsored search, thus revealing valuable – and at the same time sensitive – information regarding the distribution of user attention over the various SERP components. Despite the technical challenges that arise from this analysis, previous work has shown the utility of mouse movement patterns to measure within-content engagement [44] and predict reading experiences [45, 46]. Lagun et al. [47] introduced the concept of motifs, or frequent cursor subsequences, in the estimation of search result relevance. Similarly, Liu et al. [29] applied the motifs concept to SERPs and predicted search result utility, searcher effort, and satisfaction at a search task level. Boi et al. [48] proposed a method for predicting whether the user is looking at the content pointed by the cursor, exploiting the mouse cursor data and a segmentation of the contents in a web page. Lastly, Arapakis et al. [27, 49] investigated user engagement with direct displays on SERPs and provided further evidence that supports the utility of mouse cursor data for measuring user attention at a display-level granularity.

2.4 Inferring Emotional State

Although the connection between mouse cursor movements and the underlying psychological states has been a topic of research since the early 90s [50, 51], some studies have investigated the utility of mouse cursor data for predicting the user’s emotional state. For example, Zimmermann et al. [52] investigated the effect of induced affective states on the motor-behavior of online shoppers and found that the total duration of mouse cursor movements and the number of velocity changes were associated to the experienced arousal. Kaklauskas et al. [53] created a system that extracts physiological and motor-control parameters from mouse cursor interactions and then triangulated those with psychological data taken from self-reports, to analyse correlations with users’ emotional state and labour productivity. In a similar line, Azcarraga et al. [54] combined electroencephalography signals and mouse cursor interactions to predict self-reported emotions like frustration, interest, confidence and excitement. Yamauchi et al. [55] studied the relationship between mouse cursor trajectories and generalized anxiety in human subjects. Lastly, Kapoor et al. [56] predicted whether a user experiences frustration, using an array of affective-aware sensors.

2.5 Inferring Demographics

Yamauchi et al. [57] examined the extent to which mouse cursor movements can help identify the gender and the experienced feelings of users who were watching short film clips. Although this work provides early evidence on the utility of mouse cursor data for advanced online user profiling, it suffers from

certain limitations that we address in this work. First, the experimental setting has limited generalizability, since the adopted perception task is not very well connected to typical activities that users perform online, such as web search. Second, the data used in their predictive modeling task include multiple samples per participant randomly assigned to the training and test data partitions, hence there may be information leakage that artificially inflated model performance. In our analysis, we limit the training samples to exactly one mouse cursor trajectory per participant and test our models on unseen individuals.

Kratky et al. [58] recorded mouse cursor movements in an e-commerce website and engineered a set of meta-features to predict the user gender and age group. Their classifier was trained on several days of data per participant. Although the training and test collections had disjoint sets of participants, it was stated that the reported results were overly optimistic since researchers could not verify their ground-truth data [58]. In contrast, as discussed later, our dataset was collected from high-quality crowdworkers so we are confident that the ground-truth information is correct.

In a similar vein, Pentel et al. [59] used data from six different external sources, including e.g., keystroke data and feedback questionnaires, and hand-crafted features proposed in earlier works [35, 36, 43] to train predictive models that could identify the users' age and gender. However, because their approach relies mainly on ad-hoc data, it is less scalable and more difficult to implement than the approach we propose in this paper, which takes as input *raw* mouse cursor data. Moreover, Pentel et al. reported optimistic performance scores, which may be due to information leakage across data partitions, and omit important classification metrics such as precision, recall, and AUC. To account for their modeling approach, as well as that proposed by Kratky et al. [58], we implement the same classifier and test it in our setting.

2.6 Summary

There is a vast research literature about modeling user behavior using movement data. How we move our mouse provides a surrogate signal for gaze fixation, and therefore reveals our focus of attention, which can be used to learn *latent* interests. However there is no previous work in this regard using swipe trajectories. This is because of the lack of public datasets, which are difficult to collect in practice. Fortunately, Leiva et al. [4] have recently released a swiping dataset that enables researchers to create sophisticated Machine Learning models. To the best of our knowledge, we are the first to tap into this dataset to create different biometric-related classifiers.

3 Experiments

Based on the research literature, we hypothesize that cognitive and motor control mechanisms are embodied and reflected, to some extent, in swipe trajectories. If our classification results prove to be better than random, this work will open a new research avenue for advanced user profiling.

3.1 Dataset

The swiping dataset we used in this study is publicly available at <https://osf.io/sj67f/>. It covers both movement-level and task performance-level data, and was collected in a crowdsourcing study using a JavaScript-based virtual QWERTY *soft* keyboard. The dataset comprises 100K+ English words swiped by 1,338 users. Each word was swiped in the context of either a memorable sentence, drawn from the EnronMobile phrase set [60], or a random 4-words sentence, drawn from Google’s Trillion Word Corpus¹ and the Forbes 2019 Global 2000 list.² All words are lowercased with no punctuation symbols.

Swipe logs include the following information: event name (e.g. `touchstart`, `touchmove`), event timestamp, *x* and *y* coordinates, touch radius, and rotation angle, where available. The logs also include the keyboard size, the prompted word, and whether it was swiped correctly or not. We deliberately ignore the wrongly swiped words, as they were clear outliers, as described in the dataset [4]. There are 11,295 unique words correctly swiped which we consider for analysis. Additionally, the dataset provides the following user metadata: swipe familiarity, age, gender, nationality, browser language, device pixel ratio, screen size (height and width), English level, dominant hand, swipe hand, swipe finger, and mobile vendor. In this paper, we predict all the biometric-related variables: swipe familiarity, age, gender, nationality, English level, dominant hand, swipe hand, and swipe finger.

For the task of training our classifiers, we performed a light preprocessing on the dataset. First, we iterated through the 1,338 user logs and gathered the raw trajectories of `mousemove` *x, y* coordinates and associated timestamps *t* for each of the individual correctly swiped words. Each raw trajectory has the following format: $(x, y, t)_1, \dots, (x, y, t)_n$ where $(x, y, t)_i = (x_i, y_i, t_i)$. Then, we computed the offsets of the raw points, as it makes our sequence models scale-invariant and independent of the device’s screen size: $(\Delta x, \Delta y, \Delta t)_i = (x_i - x_{i+1}, y_i - y_{i+1}, t_i - t_{i+1})$. Eventually, our final dataset contains a total of 94,841 swiped word trajectories from 1,308 unique participants.

3.2 Predictive Targets

Figure 2 shows the class distributions of the targets we set to predict in this article. From the figures we can see that some targets are more imbalanced than others. This can potentially bias to our sequence classifiers, so we created splits of the data as balanced as possible, as explained next.

The **familiarity** target describes how well a user is used to swiping and contains 5 classes in the original dataset: “everyday” (22%), “often” (15%), “sometimes” (25%), “rarely” (27%) and “never” (11%). By looking at the class distributions, we can see that the class “often” and “never” are twice as small as the other classes. Therefore, we merged the “often” class with “everyday”

¹<https://github.com/first20hours/google-10000-english>

²<https://www.forbes.com/global2000/>

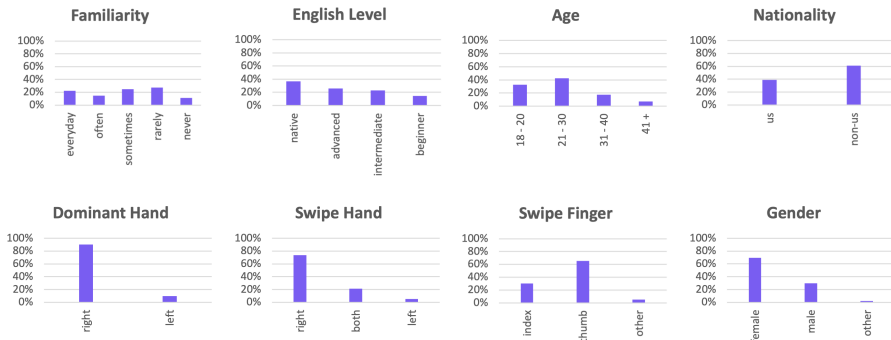
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Fig. 2: Targets and their class distributions in the original swiping dataset.

and the “never” class with “rarely”. Eventually we ended up with 3 classes for our experiments: “everyday” (37%), “sometimes” (25%) and “rarely” (38%).

The **English level** target has 4 classes in the original dataset: “native” (34%), “advanced” (27%), “intermediate” (24%), and “beginner” (15%). In this case the class distributions are fairly balanced except “beginner”, however the class semantics are important to preserve in this case. Leiva et al. [4] reported that “intermediate” and “beginner” English speakers were significantly slower and committed more errors than any of the other users, therefore we did not modify the original class distributions for the English level target.

The **age** target has 4 classes in the original dataset: “youth” (18–20 years, 33%), “young” (21–30 years, 42%), “adult” (31–40 years, 18%) and “senior” (41+ years, 7%). Since these class distributions are rather imbalanced, we redefine the age brackets as follows: “youth” (18–20 years, 33%), “young” (21–28 years, 37%), “adult” (29+ years, 30%).

The **nationality** target describes whether a user is a US citizen, given that US users were over-represented in the original dataset. The class distributions are: 39% “US” and 61% “non-US” users. Since there are two classes, we did not modify the original class distributions.

The **dominant hand** target describes the hand the user uses the most. The class distributions are: 90% “right-handed” and 10% “left-handed” users. Since there are two classes, we did not modify the original class distributions.

The **swipe hand** label describes if the user likes to use the right, left, or both hands to swipe. The original dataset has 3 classes: 74% “right hand”, 21% “both hands”, and 5% “left hand”. Due to their significant difference in size compared to the right hand class, we decided to merge the “left hand” and “both hands” classes. Eventually we ended up with 2 classes for our experiments: “right hand” (74%), “other hand” (26%).

The **swipe finger** label describes which finger the user prefers to swipe with. The original dataset has 3 classes: “index finger” (65%), “thumb” (30%), and “other finger” (5%). Given that the latter class is substantially smaller

that the rest, we left it out and consider 2 classes in our experiments: “index finger” and “thumb”.

The **gender** target includes “female” (69%), “male” (30%), and “other” (1%). Given that the latter class is substantially smaller than the rest, we left it out and consider 2 classes in our experiments: “female” and “male”.

We can see that, even after this careful class redistribution procedure, some of our targets still have notable imbalanced classes. To address this challenge we used cost-sensitive models via class weighting, as described in the next section.

3.3 Model Architectures

Since swipe trajectories are sequences of coordinates, we have chosen to train sequence models. Sequence models are Machine Learning models to handle sequential data, where each observation is dependent or contextually related to the previous one, such as text streams, audio, time-series data, etc. Recurrent Neural Networks (RNN) are by far the most popular algorithms used in sequence models. RNNs consist of standard recurrent cells, shown in Figure 3.

The typical feature of the RNN cell is a cyclic (or *loop*) connection, which enables the model to update the current state based on past states and current input data. Formally, the standard recurrent cell is defined as follows:

$$h_j = \Phi(W_h h_{j-1} + W_z z_j + b) \quad (1)$$

$$o_j = h_j \quad (2)$$

where $z_j = (x, y, t)_j$ denotes the j th vector of the input signal $\mathbf{z} = (x, y, t)_{j=1, \dots, z}$ at timestep j (i.e., the index at which a point coordinate has happened), h_j is the hidden state of the cell, and o_j denotes the cell output, respectively; W_h and W_z are the weight matrices; b is the bias of the neurons; and Φ is an activation function.

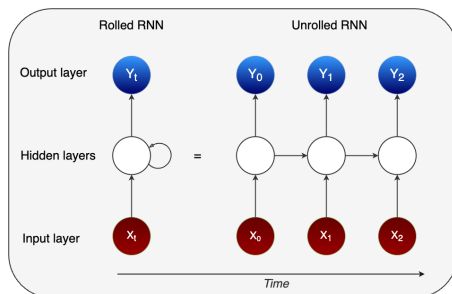


Fig. 3: Standard RNN architecture.

Standard recurrent cells have achieved success in many sequence learning problems such as natural language processing [61], action recognition [62], or

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image captioning [63]. However, the standard recurrent cells are not capable of handling long-term dependencies. To solve this issue, the LSTM and GRU cells were developed [64, 65]. They improve the capacity of the standard recurrent cell by introducing different gates, which we briefly describe as follows.

On the one hand, the LSTM cell is defined as follows:

$$G_i = \sigma(W_u[h_{j-1}, z_j] + b_i) \quad (3)$$

$$G_f = \sigma(W_f[h_{j-1}, z_j] + b_f) \quad (4)$$

$$G_o = \sigma(W_o[h_{j-1}, z_j] + b_o) \quad (5)$$

$$c_j = G_i \odot \tilde{c}_j + G_f \odot c_{j-1} \quad (6)$$

$$\tilde{c}_j = \Psi(W_c[h_{j-1}, z_j] + b_c) \quad (7)$$

$$h_j = G_o \odot \Psi(c_j) \quad (8)$$

where c_j is a hidden state responsible for the long-term memory, $\tilde{c}_j$ is a candidate state responsible for controlling the cell-state data, W_* are weight matrices, b_* are biases, G_* denote cell gates (i: input, f: forget, o: output), and Ψ and σ are activation functions. The $\odot$ operator denotes the Hadamard (element-wise) product.

As noted, the LSTM has two kinds of hidden states [66]: a “slow” state c_j that keeps long-term memory, and a “fast” state h_j that makes decisions over short periods of time. The forget gate G_f can decide what information will be thrown away from the cell state. In practice, the activation functions Ψ and σ are hyperbolic tangent and sigmoid, respectively, however other non-linear functions have been promoted in the research literature; see the next section for a brief discussion.

On the other hand, the GRU cell is defined as follows:

$$G_u = \sigma(W_u[h_{j-1}, z_j] + b_u) \quad (9)$$

$$G_r = \sigma(W_r[h_{j-1}, z_j] + b_r) \quad (10)$$

$$\tilde{c}_j = \Psi(W_c[G_r \odot h_{j-1}, z_j] + b_c) \quad (11)$$

$$h_j = G_u \odot \tilde{c}_j + (1 - G_u) \odot h_{j-1} \quad (12)$$

where the forget and input gates of the LSTM cell are now combined into a single update gate G_u . A reset gate G_r controls which parts of the state get used to compute next cell state. As it can be observed, GRU controls the flow of information like the LSTM cell, but without having to use memory (the forget and output gates in LSTM): the full state vector is output at every time step. Furthermore, performance-wise the GRU cell is on par with the LSTM in many problems [67, 68] but it is computationally more efficient because of its less complex structure. Therefore, our classifier uses GRU cells for its RNN layer.

On the other hand, the GRU cell is defined as follows:

$$G_u = \sigma(W_u[h_{j-1}, z_j] + b_u) \quad (13)$$

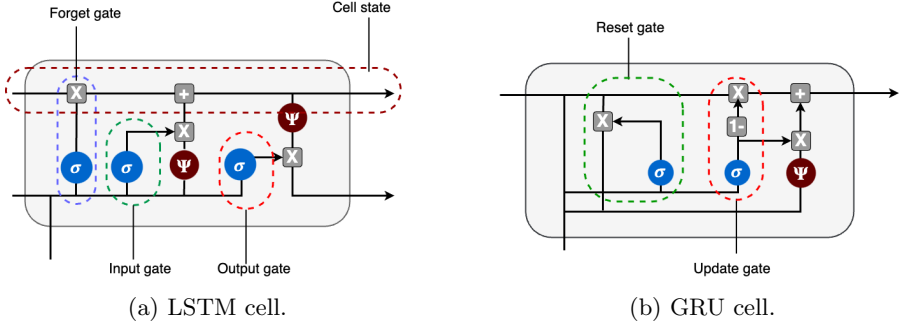


Fig. 4: Overview of the LSTM and GRU cells.

$$G_r = \sigma(W_r[h_{j-1}, z_j] + b_r) \quad (14)$$

$$\tilde{c}_j = \Psi(W_c[G_r \odot h_{j-1}, z_j] + b_c) \quad (15)$$

$$h_j = G_u \odot \tilde{c}_j + (1 - G_u) \odot h_{j-1} \quad (16)$$

where the forget and input gates of the LSTM cell are now combined into a single update gate G_u . A reset gate G_r controls which parts of the state get used to compute next cell state. The LSTM and GRU architectures are presented in figure [Figure 4](#).

As can be observed, GRU controls the flow of information like the LSTM cell, but without having to use memory (the forget and output gates in LSTM): the full state vector is output at every time step. Furthermore, performance-wise the GRU cell has been shown to be on par with the LSTM in many problems [\[67, 68\]](#) while at the same time being computationally more efficient because of its less complex structure. It remains unclear, however, what would be the best RNN-based architecture configuration (e.g. type of cell, number of layers, number of neurons, etc.) for predicting demographic and behavioral correlates from swipe trajectories. Therefore, we study what is the most suitable configuration.

The hyperbolic tangent activation function is applied in every recurrent layer, since recurrent neural networks do not only get gradients from lower layers but also from previous timesteps. The derivatives of tanh are larger than the other activation functions such as sigmoid or swish. Thus, the cost function is minimized faster by using the hyperbolic tangent activation function. It has a normalized range from -1 to 1 and the output is symmetric around 0 which leads to faster convergence. Formally:

$$\tanh(z) = \frac{e^z - e^{-z}}{e^z + e^{-z}} \quad (17)$$

Finally, the classification rule of all the classifiers we trained is defined by:

$$\hat{z} = \text{softmax}(W_z \odot h_j + b_z) \quad (18)$$

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13
14 where $\text{softmax}(\mathbf{v})$ is the softmax function, which normalizes the output of the
15 last network layer (denoted as $\mathbf{v}$ for the sake of simplicity) to a probability
16 distribution over the predicted output classes K :

$$17 \quad \text{softmax}(\mathbf{v})_i = \frac{e^{v_i}}{\sum_{j=1}^K e^{v_j}} \quad (19)$$

18
19 for $i = \{1, \dots, K\}$ and $\mathbf{v} = (v_1, \dots, v_K)$.

24 **3.4 Methodology**

25
26 Neural networks require inputs with the same shape and size, at least within
27 the same batch. However, the trajectories in the swiping dataset have varying
28 sequence lengths (**Mean=72, StDev=64**). Thus, we need to apply padding and
29 truncation to the sequence trajectories in order to ensure equally-sized lengths.
30 We chose 200 points as our maximum sequence length, based on the data
31 distribution, thus all sequences were padded or truncated to this maximum
32 capacity.

33
34 For the data split ratio, we **randomly** used 60% for the training set, 20% for
35 the validation set and the remaining 20% for the testing set. **Each model was**
36 **trained and evaluated on the very same data partitions. The testing partition**
37 **is held out exclusively for model evaluation, as it represents unseen data and**
38 **is therefore a more challenging but also a more realistic scenario.** We chose to
39 work with the Adam optimizer since it is an algorithm for efficient stochastic
40 optimization that only requires first-order gradients with little memory require-
41 ment [69]. The loss function to minimize is sparse categorical cross-entropy,
42 since our task is a multi-class classification problem with $C \geq 2$ classes.

45 **3.4.1 Model Hyperparameters**

46
47 We conducted a systematic analysis regarding which architecture hyperparam-
48 eters affect model performance. The architecture choices include the type of
49 layer (RNN, LSTM or GRU), whether bidirectional or not, number of hidden
50 neurons, number of hidden layers, and optimizer's learning rate.

51
52 In bidirectional models, the input flows in both directions such that every
53 component of an input sequence is able to use the information from both
54 the past and present. The model adds for each recurrent layer another recur-
55 rent layer with the reversed information flow. The outputs from forward and
56 backward layers are combined by concatenating them.

57
58 The learning rate controls how quickly a neural network model adapts to
59 a problem. By choosing smaller learning rates, the model might result in a
60 long training or might get stuck in the process since the changes made to
61 the weights each update are smaller. On the other hand, the model might
62 converge too fast to a sub-optimal solution by choosing larger learning rates.
63 We experimented with learning rate (LR) 10^{-1} , 10^{-2} , 10^{-3} , 10^{-4} and 10^{-5} , 5

variations in total. Note that LR higher than 10^{-1} is unlikely to work because of the Robbins-Monro condition: the more LR approaches 1, the less room for the model to learn from backpropagation. Further, when $LR > 1$, stochastic gradient descent is not guaranteed to converge to a minimum [70].

We tested the performance of the different layer types (RNN, BiRNN, LSTM, BiLSTM, GRU and BiGRU), 6 variations in total for each target. The number of hidden layers we tested range from 1 to 4 layers in our model architectures, 4 variations in total. The number of hidden units range from 10 to 150, with a step size of 10, 15 variations in total. Eventually, we tested 30 different hyperparameter variations for each predicted target, totalling a number of 240 different trained classifiers.

3.4.2 Model Regularization

We use two popular regularization techniques to prevent overfitting and make our classifiers more generalizable: Dropout and Early Stopping. Dropout removes neurons at random during training, to force different connection paths between neurons and thus avoid the model to take shortcuts (like creating spurious correlations between inputs and outputs). We used Dropout with 0.15 probability in all RNN layers. Early Stopping stops training when no further improvements over a given metric (e.g. classification accuracy, validation loss) are made. We set 40 epochs of patience for Early Stopping while monitoring the F1 score, i.e. if the model does not improve the F1 score over the validation data in 40 consecutive epochs, training stops and the best model weights are retained. The maximum number of training epochs is set to 500 and batch size is set to 32 trajectories.

As discussed in the previous section, our class distributions are imbalanced for most of the targets. To address this challenge we used cost-sensitive models via class weighting. The idea is to give all classes equal importance on gradient updates, regardless of how many samples we have from each class in the training data. This prevents models from predicting the more frequent class more often just because it is more common. Instead, class weighting forces models to learn a more apt representation of each class.

3.4.3 Evaluation Metrics

To evaluate model performance, we should note that classification accuracy is not the best metrics to report, since our data is not perfectly balanced. Therefore, we chose the F1 score as our main evaluation metric, since it very popular for imbalanced classification problems. The F1 score (Equation 22) is the harmonic mean of Precision and Recall. Precision (Equation 20) is the ratio of correctly predicted positive instances (True Positives, TP) divided by the total number of predicted positive instances both True (TP) and False Positives (FP). Recall (Equation 21) identifies the number of correct positive predictions among all positive predictions that were possible, including TP and False Negatives (FN). We also report the Area Under the ROC curve (AUC), which determines the discriminatory power of any classifier [71].

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad (20)$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (21)$$

$$\text{F1 score} = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (22)$$

4 Results and Discussion

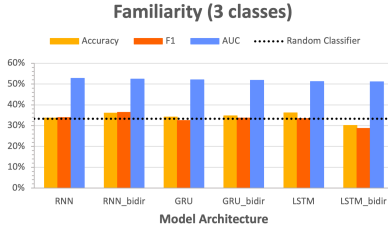
In the following we present the results obtained in our experiments. To test our research hypothesis, we also consider the accuracy of a random classifier as baseline, denoted with a horizontal black dotted line in all plots, for which we compute the *a priori* distribution of the number of classes C , i.e.

$$\text{Random Accuracy (\%)} = \frac{100}{C} \quad (23)$$

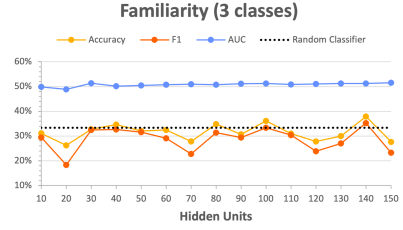
As explained in previous sections, if the models we train are no better than a random classifier, then we must reject our research hypothesis and conclude that swiping data do not reveal demographic or behavioral information about the user. Otherwise, we must validate our research hypothesis and report the best model configuration for each predicted target.

In our experiments we have investigated how model performance metrics vary by manipulating the different hyperparameters: layer type (RNN, GRU, LSTM), bidirectionality, number of hidden layers and hidden units, and optimizer's learning rate. The best performance results obtained through a systematic analysis of the chosen hyperparameters are reported from [Figure 5](#) to [Figure 12](#). As can be observed, in all cases, the trained models' performance is better than that of a random classifier, which indicates that our models are able to learn latent representations about the demographic and behavioral information from swiping trajectories.

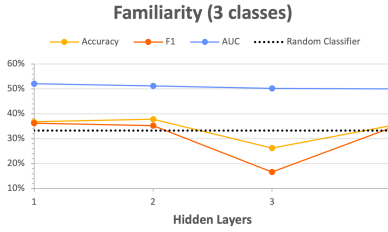
[Table 1](#) summarizes the best performing models and their corresponding hyperparameters. We also provide a random classifier baseline to contextualize the importance of the results. As shown in the table, the classifiers performed best on most demographic and behavioural correlates by defining the model to be bidirectional and choosing recurrent layers with LSTM or GRU cells. All models performed better than a random classifier, with an increase from 4% to 10% for the swiping familiarity, age and swiping hand targets, and with an increase from 10 to almost 40% for the English level, nationality, gender, dominant hand and swiping finger targets. Overall, our results show that the trained classifiers were able to identify latent demographic traits from the swiping trajectories, hence validating our hypothesis that swiping carries rich information about the user. This opens interesting research avenues which are yet to be fully explored. For example, new techniques to prevent advanced user profiling should be devised in future work.



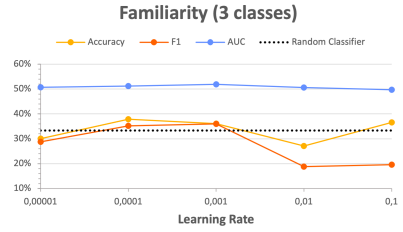
(a) Layer type



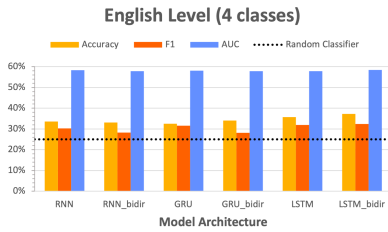
(b) Hidden Units



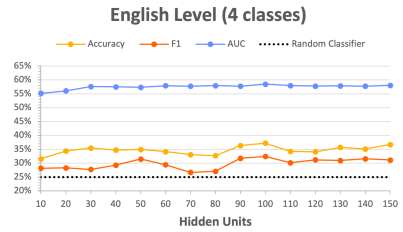
(c) Hidden Layers



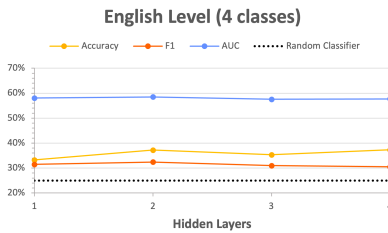
(d) Learning Rate

Fig. 5: Predicting the user's swiping familiarity from swiped words trajectories.

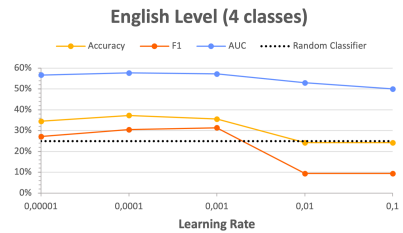
(a) Layer type



(b) Hidden Units

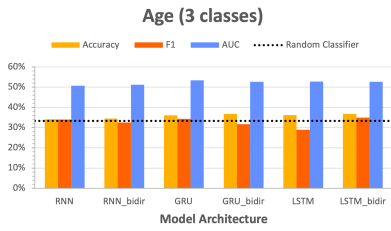


(c) Hidden Layers

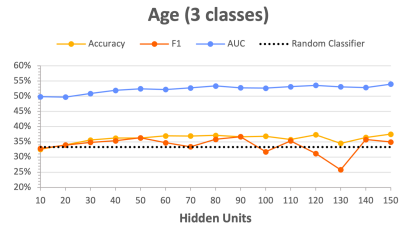


(d) Learning Rate

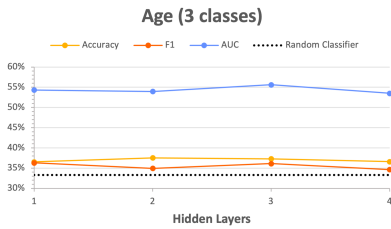
Fig. 6: Predicting the user's English level from swiped words trajectories.

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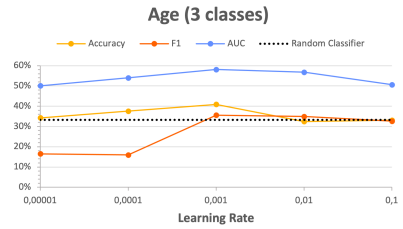
(a) Layer type



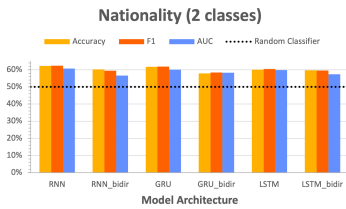
(b) Hidden Units



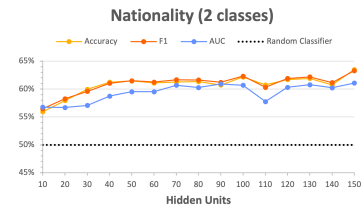
(c) Hidden Layers



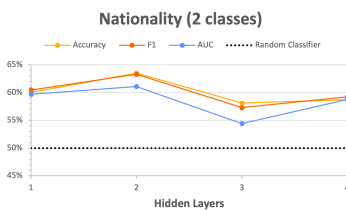
(d) Learning Rate

Fig. 7: Predicting the user's age from swiped words trajectories.

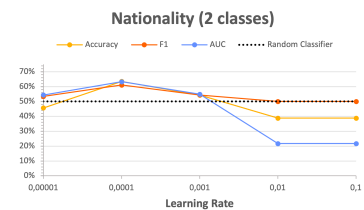
(a) Layer type



(b) Hidden Units

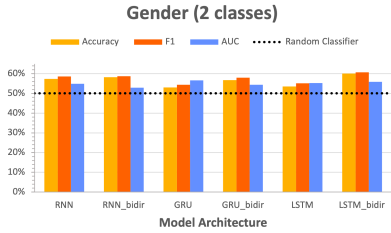


(c) Hidden Layers

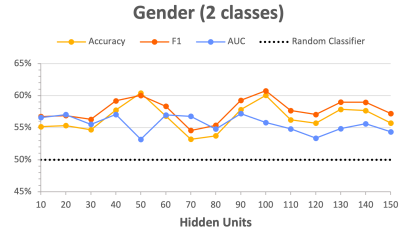


(d) Learning Rate

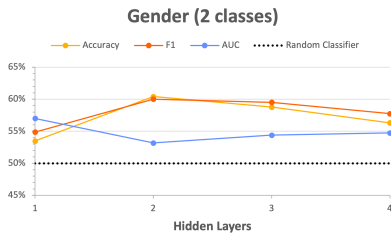
Fig. 8: Predicting the user's nationality from swiped words trajectories.



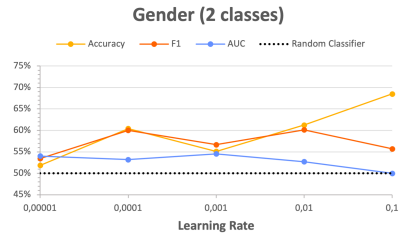
(a) Layer type



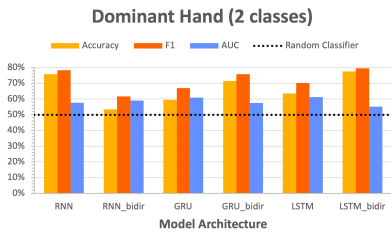
(b) Hidden Units



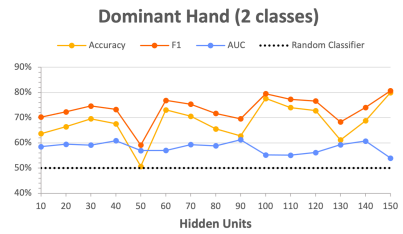
(c) Hidden Layers



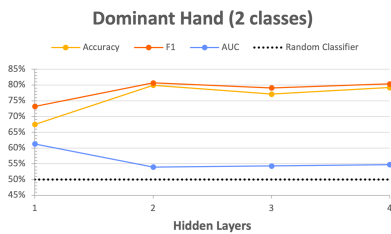
(d) Learning Rate

Fig. 9: Predicting the user's gender from swiped words trajectories.

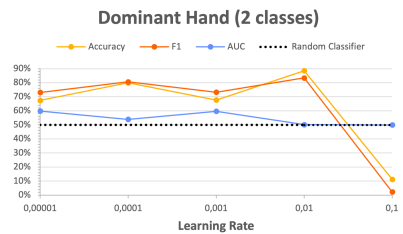
(a) Layer type



(b) Hidden Units



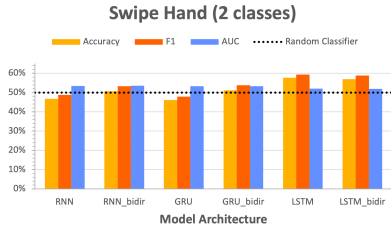
(c) Hidden Layers



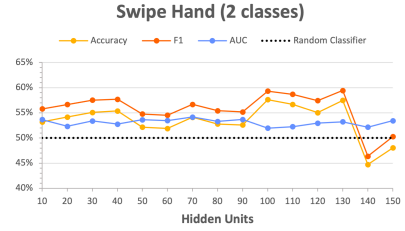
(d) Learning Rate

Fig. 10: Predicting the user's dominant hand from swiped words trajectories.

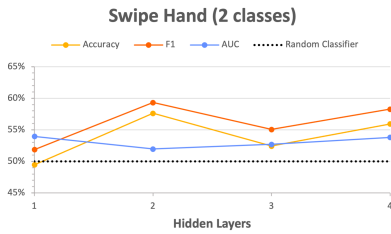
What Can a Swiped Word Tell Us More?



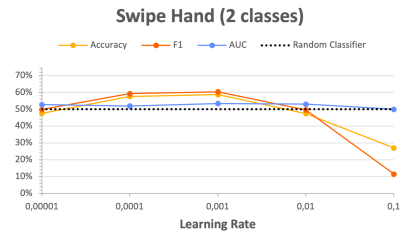
(a) Layer type



(b) Hidden Units

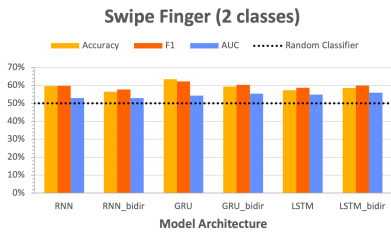


(c) Hidden Layers

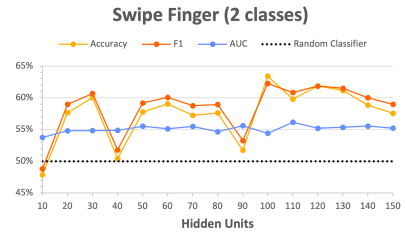


(d) Learning Rate

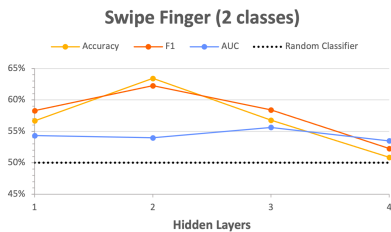
Fig. 11: Predicting the user's swiping hand from swiped words trajectories.



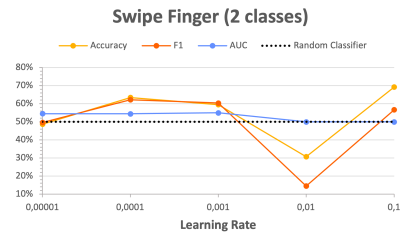
(a) Layer type



(b) Hidden Units



(c) Hidden Layers



(d) Learning Rate

Fig. 12: Predicting the user's swiping finger from swiped words trajectories.

Target	Model	Layers	Units	LR	Rand. Acc.	Model Acc.
Age	BiGRU	2	150	10^{-3}	33.33%	40.91%
Gender	BiLSTM	2	50	10^{-1}	50%	68.51%
Nationality	RNN	2	150	10^{-4}	50%	63.50%
Familiarity	LSTM	2	140	10^{-4}	33.33%	37.87%
English level	BiLSTM	4	100	10^{-4}	25%	37.32%
Dominant hand	BiLSTM	2	150	10^{-2}	50%	88.57%
Swiping hand	LSTM	2	100	10^{-3}	50%	58.74%
Swiping finger	GRU	2	100	10^{-1}	50%	69.28%

Table 1: Summary of the best model architectures for predicting demographic (top rows) and behavioral (bottom rows) information from swiping trajectories.

Finally, we also report the confusion matrices for each of the best performing models in Figure 13. As we can see, in most cases the trained models were able to successfully discriminate between classes. However, in other cases the models failed to learn properly to discriminate and degenerated as ZeroR classifier, i.e. a model that always predicts the majority class. This only happened to the models that predicted gender, dominant hand, and swiping finger. In any case, the performance of all models is better than that of a random classifier, therefore we validate our research hypothesis and conclude that, overall, demographic and behavioral information can be inferred from swiping trajectories.

5 Limitations and Future Work

In this work we have focused on single-word classification, which is the most challenging scenario since some words might be as short as two characters and therefore the information a classifier can infer is minimal. It would be interesting to predict demographic and behavioral attributes at the sentence level, i.e., by considering all swiped words a user might enter in a row. This could be done, e.g., by concatenating the feature vectors of each word in the sentence. In addition, we have not considered word difficulty in our analysis. This might be an interesting avenue for future work, since it has been shown that experienced shape-writing users tend to be more loose while entering non-familiar words [4]. It may be the case, therefore, that other demographic and behavioral traits are highly correlated with swiping patterns. For example, ageing is marked by a decline in motor control abilities, which is reflected by the users' writing performance. Smith et al. [72] observed that older people incurred in longer movement times, but also more sub-movements and more pointing errors than the young. Unfortunately, we cannot study this effect in the dataset we have analyzed, since most users are aged between 20 and

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Fig. 13: Confusion matrices for the best performing models reported in Table 1.

40 years. More concretely, the dataset is skewed towards young female right-handed participants.

Our sequence models are *relatively* costly to train because of their recurrent nature and the large amount of swiping data we analyzed. For example,

training the non-bidirectional LSTM models took between 1.5 to 10 hours in the High-Performance Computing facilities of our university, whereas the same model with bidirectional layers took between 3 and 20 hours. While these training times might be small for today’s standards, real-time processing is clear out of reach at present, which means that we cannot build our models as new data come in. We also have not experimented with attention mechanisms such as local [73] and global [74] attention. Attention has been proved to be effective for recurrent models, but we decided to go for an initial exploration of the feasibility of our approach in the first place. Given that our results are promising, we can even experiment with other architectures such as Transformers in future work.

Furthermore, we have explored sequential models only, as they are better suited to handle the sequential nature of *swipe* trajectories. Observing the promising results obtained in our study, we believe that other data representations should be explored. For example, if we consider images (pixel values) instead of discrete spatio-temporal time series as input data, then we could try Convolutional Neural Networks (CNNs) as well as more powerful architectures like Vision Transformers (ViT). This is left as an interesting opportunity for future work.

Another interesting opportunity for future work is to consider more balanced splits of the data. Given the number of target classes, there are 1152 possible combinations, not all of which may be well represented, and are certainly not represented equally. We should point out, however, that our study is the first to use the original “How we Swipe” dataset for training Machine Learning models and to get a more realistic estimate of their performance. We also have not considered any interactions between targets, such as gender and age. For example, training only for young females (which happens to be the most represented group of participants) might result in more accurate models.

Finally, we should remark that all mobile vendors provide a swiping service to millions of users. Although, they ask for consent to track swiping information for the purpose of improving the service and enhancing user experience, our study brings to light that extracting personal information of users from swiping data is also within reach. The significantly large amount of swiping data mobile vendors can have access to, allows them to build a powerful demographic and behavioral inference engines. Thus, the users can potentially give up their privacy, without actually consenting to it, only by sharing their swiping information with mobile vendors.

6 Conclusion

We have analyzed the dynamics of swiping trajectories from the only publicly available swiping dataset to model and predict different demographic and behavioural correlates. We explored three popular sequence model architectures (RNN, LSTM, GRU) and conducted a systematic analysis regarding

which model hyperparameters yield better classification performance. Overall, the classifiers performed best on most demographic and behavioural traits by defining the models to be bidirectional and choosing recurrent layers with LSTM or GRU cells having 100 units each. We noticed that the eight considered targets are challenging to predict, however the results from our experimental evaluations demonstrate that all models perform better than a random classifier, with an increase from 4 to 10% for the swiping familiarity, age, and swiping hand targets, and with an increase from 10 to 40% for the English level, nationality, gender, dominant hand, and swiping finger targets. For the three latter targets (gender, dominant hand, and swiping finger) the trained models tended to predict the majority class.

Taken together, our results show that the classifiers are able to identify latent demographic and behavioral information from swiping trajectories, hence validating our research hypothesis that swiping carries rich information about the user. This finding may have unexpected consequences for user's privacy, since currently swiping is supported by all mobile vendors and has millions of users, so people may be inadvertently profiled at an unprecedented granularity. It is a matter of time for researchers to build more sophisticated Machine Learning models, therefore future work should consider new ways of addressing these issues without impacting the user's swiping experience.

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Declarations

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Competing interests

The authors have no competing interests to declare that are relevant to the content of this article.

Availability of data and materials

The swiping dataset we used is publicly available at <https://osf.io/sj67f/>.

Authors' contributions

D.C.A. Lemarquis: Software, Writing - Original Draft. **B.A. Yilma:** Methodology, Writing - Original Draft. **L.A. Leiva:** Conceptualization, Methodology, Writing - Reviewing and Editing.

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